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Display units with automated power governing

Abstract

Display units for prioritizing requested operational updates are disclosed. A control subsystem is in electronic communication with an electronic display, sensor(s), and a thermal management subsystem. The control subsystem operates the display unit based on operational parameters which are specific to readings from the one or more sensors. Requests to update operational parameters for the display unit are received at the control subsystem and are prioritized for implementation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/885,160 filed Sep. 13, 2024, which is a continuation of U.S. application Ser. No. 18/652,287 filed May 1, 2024, now U.S. Pat. No. 12,118,953 issued Oct. 15, 2024, which is a continuation of U.S. application Ser. No. 18/214,948 filed Jun. 27, 2023, now U.S. Pat. No. 12,027,132 issued Jul. 2, 2024, the disclosures of which are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

(1) Exemplary embodiments relate generally to display units with automated power governing as well as systems and methods related to the same.

BACKGROUND AND SUMMARY OF THE INVENTION

(2) Digital out of home advertising has grown in recent years, and continues to be a major source of interest, not only for advertising, but also for other public announcements, various marketing and

other services, “smart city” services, telecommunication services, and the like. These units are exposed to a wide variety of operating conditions (e.g., weather, user, and/or environment based) and demands (e.g., usage and/or programmed operational parameters). It is known to provide electronic displays in ruggedized enclosures, such as with thermal management and/or remote monitoring and/or control functions, to provide survivability and adaptability of such units under such demanding conditions. For example, it is known to limit alternating current (AC) draw on various electronic devices. It is also known to limit direct current (DC) draw on various electronic devices. It is also known to set thermal limits on various electronic devices where conditions are changed in response to meeting the thermal limit. However, current solutions fail to gracefully control such units in a fashion which exerts sufficient automated control to prevent or limit disruptions to user experiences while also safeguarding units from failure. What is needed is a power governing control system which reliably and operably controls such units to reduce or prevent failure while also minimizing disruption to user experience.

(3) Units with power governing control systems which reliably control such units to reduce or prevent failure while also minimizing disruption to user experience are provided, along with systems and methods related to the same. Units may include one or more electronic displays. The units may include a control system with one or more of: an AC governor, a DC governor, and a thermal governor. The control system may be electrically interposed between a power source (e.g., external utility power supply) for the unit and some or all electricity consuming components of the unit (e.g., electronic displays, thermal management systems, customer equipment, peripheral equipment, combinations thereof, or the like). The governors may operably control illumination sources for the electronic displays, such as the backlights.

(4) For the AC governor, where an AC current threshold is met or exceeded, the AC governor may operate in an AC current mitigation mode, such as by reducing power supplied to the illumination sources and/or other electricity consuming components rapidly, such as in a matter of seconds, and/or less than one second. The AC current threshold may be set relative to service rated current, such as between 70%-99% of the service rating. Power levels may be automatically increased where the AC current threshold is no longer met.

(5) One or more circuit breakers or the like may be electrically interposed between the power source for the unit and some or all electricity consuming components of the unit to serve as a backup in case of continued AC current increase. The AC current threshold may be set to below the circuit breaker ratings, such as at 70%-99% thereof. AC current input may be monitored by the control system and electronic notifications may be generated and/or transmitted to remote device(s) where the AC current input is below a predetermined threshold, such as an expected current input or a margin thereof.

(6) For the DC governor, where a DC current demand threshold is met or exceeded, the DC governor may operate in a DC current mitigation mode such as by reducing, at a relatively moderate rate, such as in a matter of multiple seconds or minutes, power to the illumination sources and/or other electricity consuming components, such as down to a zero level. Power levels may be automatically increased where the DC current threshold is no longer met.

(7) Where one or more service limits of DC power supplies are met or exceeded, which may be a threshold above the DC current demand threshold, the DC power supplies may be automatically shut off and begin a restart sequence.

(8) For the thermal governor, where the thermal management system of a unit is at maximum capacity or some other threshold capacity (e.g., 70-99% thereof), and any one or more internal temperatures, such as measuring by one or more temperature sensors in electronic communication with the control system, the thermal governor may operate in a thermal mitigation mode, such as by reducing to relatively slow, such as over a matter of several minutes, reduce power provided to the illumination sources, such as down to a zero level. Power levels may be automatically increased where the DC current threshold is no longer met, such as over a period of a same or different

number of minutes.

(9) The power adjustments shown and/or described herein, such as by the AC, DC, and/or thermal governors may be provided on various bases, such as but not limited to, on a linear, exponential and/or the like basis relative to the respective threshold(s). A grace value may be set such that the governors are configured to remain within the various mitigation modes unless/until the relevant values pass the respective thresholds by at least a predetermined amount. This may reduce or prevent rapid and frequent transitions between the normal mode and mitigation mode(s).

(10) The governors may be operated independently and may independently provide benefits for reducing or eliminating failures and interruptions to user experiences. Analysis of measures against thresholds for each governor may be performed in parallel or in any sequence. The combination and operation of the governors, in particular, may provide exceptional reduction or elimination of failures and interruptions to user experiences. The integrated safety feature may alternatively, or additionally, permit the reduction in power supplies for a unit and/or reduce power consumption for a unit, among other benefits. This may allow provided power supplies to operate closer to capacity, whereby increased efficiencies are generally found. A centralized control system may prioritize received requests from the governor(s) for operational changes, such as but not limited to, by prioritizing the largest reductions to illumination levels.

(11) Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

(2) FIG. 1 is a simplified plan view of an exemplary unit with automated power governing features and related system components;

(3) FIG. 2 is a simplified perspective view of the unit of FIG. 1 in an exemplary environmental operation context;

(4) FIG. 3 is a flow chart with exemplary logic for operating the unit of FIG. 1, such as when operating in the various environmental operation conditions illustrated by FIG. 2; and

(5) FIG. 4 is a flow chart with other exemplary logic for operating the unit of FIG. 1, such as when operating in the various environmental operation conditions illustrated by FIG. 2 and/or in conjunction with the logic of FIG. 3.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

(6) Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

(7) Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular

shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(8) FIG. 1 illustrates an exemplary outdoor display unit **10** (hereinafter “unit” for brevity). The unit **10** may include one or more electronic displays **12** (hereinafter also “display” for brevity). Each electronic display **12** may include liquid crystal displays (LCDs), organic light emitting diode displays (OLED), light emitting diode (LED) displays, plasma displays, cathode ray tube displays, rear projection displays, combinations thereof, or the like. Each electronic display **12** may include one or more backlights, such as direct backlights, edge lighting, combinations thereof, or the like. Each display **12** may not require a separate backlight, such as in the case of OLEDs, which are self-illuminating.

(9) The units **10** may include one or more thermal management systems **20**. The thermal management systems **20** may comprise one or more fans, open loop airflow pathways, closed loop airflow pathways, thermoelectric modules, air conditioning units, sensors **24**, combinations thereof, or the like. The thermal management systems **20** may be, for example, without limitation, as shown and/or described in one or more of: U.S. Pat. No. 9,629,287 granted Apr. 18, 2017 entitled SYSTEM FOR USING CONSTRICTED CONVECTION WITH CLOSED LOOP COOLING SYSTEM AS THE CONVECTION PLATE, U.S. Pat. No. 10,506,738 granted Dec. 10, 2019 entitled CONSTRICTED CONVECTION COOLING FOR AN ELECTRONIC DISPLAY, U.S. Pat. No. 11,540,418 granted Dec. 27, 2022 entitled ELECTRONIC DISPLAY WITH COOLING, and/or U.S. Pat. No. 11,032,923 granted Jun. 8, 2021 entitled FIELD SERVICEABLE DISPLAY ASSEMBLY, the disclosures of each of the foregoing being hereby incorporated by reference as if fully restated herein. Other types and/or kinds of thermal management systems **20** may be utilized.

(10) The units **10** may include one or more additional electricity consuming components **22**, such as but not limited to, radio transmitters/receivers (e.g., “5G” wireless equipment), cameras, touchscreens, sensors, servers, sensors **24**, combinations thereof, or the like.

(11) The units **10** may comprise one or more sensors **24**. The sensors **24** may comprise temperature sensors, pressure sensors, air quality sensors, air flow sensors, location detection devices, light sensors, color sensors, combinations thereof, or the like. Any number and/or type of sensors **24** may be provided at any number of locations within the units **10**.

(12) The control subsystem **14** may be configured to adjust electronic display **12** illumination (e.g., backlight, OLED) based on sensed ambient lighting conditions, such as by way of the one or more sensors **24**. In exemplary embodiments, without limitation, the control subsystem **14** may be configured to normally increase the power level of the electronic displays **12** under relatively high ambient light conditions to provide increased image visibility. In exemplary embodiments, without limitation, the control subsystem **14** may be configured to normally decrease the power level of the electronic displays **12** under relatively low ambient light conditions to save power. Such components and/or operations may be, for example, without limitation, as shown and/or described in U.S. Pat. No. 10,440,790 granted Oct. 8, 2019 entitled ELECTRONIC DISPLAY SYSTEM WITH ILLUMINATION CONTROL, the disclosures of which are hereby incorporated by reference as if fully restated herein. Other components and/or methods of control may be utilized.

(13) The electronic displays **12**, thermal management systems **20**, additional electricity consuming components **22**, sensors **24**, combinations thereof, or the like may be in electronic communication with a control subsystem **14**, which may receive data therefrom and/or provide operational commands to such components.

(14) The control subsystem **14** may, additionally or alternatively, be in electronic communication with a power subsystem **16**. The power subsystem **16** may comprise one or more power supplies (e.g., DC power supplies), power transformers, power regulation components, capacitors, bulk energy storage devices (e.g., batteries), power converters (e.g., AC/DC converters), switches, combinations thereof, or the like. The power subsystem **16** may be electrically interposed between some or all electricity consuming components of the unit **10** (e.g., electronic displays **12**, thermal

management systems **20**, additional electricity consuming components **22**, sensors **24**, combinations thereof, or the like) and a power source **26** (e.g., external utility power supply).

(15) Alternatively, or additionally, the control subsystem **14** may be electrically interposed between some or all electricity consuming components of the unit **10** (e.g., electronic displays **12**, thermal management systems **20**, additional electricity consuming components **22**, sensors **24**, combinations thereof, or the like) and the power source **26**. The power source **26** may comprise one or more external utility power supplies, such as power generation facilities (e.g., power plants), utility lines, and/or power grids, bulk energy storage devices (e.g., batteries), local power supplies (e.g., wind power, portable or fixed power generators, solar power, combinations thereof, or the like), combinations thereof, or the like. One or more power sources **26** may be available (e.g., solar power with backup utility power). The power sources **26** may be external to the units **10**. In at least the case of solar power and/or bulk energy storage devices, some or all of the power sources **26**, or components thereof, may be, in whole or in part, internal to the unit **10**.

(16) Alternatively, or additionally, a power overload prevention subsystem **18** may be electrically interposed between some or all electricity consuming components of the unit **10** (e.g., electronic displays **12**, thermal management systems **20**, additional electricity consuming components **22**, sensors **24**, combinations thereof, or the like) and the power source **26**. The power overload prevention subsystem **18** may comprise one or more circuit breakers, fuses, combinations thereof, or the like.

(17) The control subsystem **14** may comprise, or be in electronic communication with (such as, by way of non-limiting example, provided at the power subsystem **16** or otherwise) one or more of: an AC power governor **28**, DC power governor **30**, and thermal governor **32** (collectively the “governors” for brevity). One or more of the governors may be independent components, part of the control subsystem **14**, part of the power subsystem **16**, combinations thereof, or the like. One or more of the governors may comprise hardware components, such as but not limited to, processors, electronic storage devices, computing devices, switches, power transformers, power limiters, power regulators, combinations thereof, or the like, and/or software components, such as but not limited to, software code, variables, algorithms, operational command subroutines, combinations thereof, or the like.

(18) The control subsystem **14**, power subsystem **16**, and one or more of the governors may comprise one or more power meters and/or simulated power meters, such as shown and/or described in one or more of, and/or use one or more of the techniques shown and/or described in: U.S. Pat. No. 11,022,635 granted Jun. 1, 2021 entitled MEASURING POWER CONSUMPTION OF AN ELECTRONIC DISPLAY ASSEMBLY. Such power meters and/or simulated power meters and/or related techniques may serve as sensor(s) **24**, though such is not required.

(19) The control subsystem **14** may comprise, or be in electronic communication with, one or more network communication devices, such as for wired and/or wireless transmission and/or receipt of data. Such data may include data regarding unit **10** operation, commands, protocols, software, and/or thresholds, combinations thereof, or the like. The network communication devices may be configured to facilitate electronic communication by way of one or more internets, intranets, cellular networks, combinations thereof, or the like.

(20) The units **10** may comprise one or more internal support frameworks, external housings, cover panels, ventilation systems, filters, openings, combinations thereof, or the like, and may be provided in a variety of sizes, shapes, and/or configurations. Some or all of the components of the units **10** shown and/or described may be internal to the units **10** and/or external thereto.

(21) As illustrated with particular regard to FIG. 2, a large variety of ambient weather and other operating conditions experienced by the units **10** may result in drastic fluctuations to unit **10** needs and/or operational characteristics, such as thermal management subsystem **20** demands, experienced temperatures (e.g., due to ambient temperatures, solar loading, solar angle, seasonal changes, weather patterns, combinations thereof, or the like), ambient lighting conditions (e.g., sun

and moonlight movement, shade, cloud cover, precipitation, seasonal changes, combinations thereof, and the like) and resulting need to alter electronic display **12** illumination levels to render image sufficiently visible, combinations thereof, or the like. By way of non-limiting example, relatively warm temperatures and/or high solar loading may increase demand on the thermal management subsystem **20** for cooling and/or desire for increased illumination levels of the electronic displays **12** for image visibility, resulting in relatively high electrical demand on the unit **10** and/or high internal temperature levels, thereby pushing the unit **10** to the extremes of its operating abilities. Changing global weather patterns and extreme weather events may contribute to such extreme conditions. The automated power governing features shown and/or described herein may assist with managing the unit **10** such that the most extreme of such conditions, reducing the need to otherwise significantly engineer the units **10** for such unlikely or rare events. This may reduce design complexity and unit **10** cost while improving reliability and user experience. The illustrated and/or described examples of operating conditions are merely exemplary and not intended to be limiting.

(22) FIG. **3** provides exemplary logic for operating the unit **10**, such as during the various exemplary operating conditions of FIG. **2**. The control subsystem **14** may operate the unit **10** normally. Such normal operations may include, for example, without limitation, displaying content at the electronic displays **12**, adjusting illumination of the electronic displays **12**, such as in response to ambient light readings taken by one or more of the sensors **24**, operating the thermal management subsystem **20**, such as by running fans at various speeds based on temperature readings taken by one or more of the sensors **24**, operating customer or peripheral equipment, such as providing wayfinding information, telephonic, voice, and/or video calls, operating wireless “hotspots”, receiving and/or processing information from the additional electronic equipment **22**, such as cameras, combinations thereof, or the like. These are merely exemplary and not intended to be limiting.

(23) While undertaking normal operations, the control subsystem **14** may monitor one or more operating conditions of the unit **10**, such as by way of the sensors **24**, thermal management subsystem **20**, and/or power subsystem **16**. In exemplary embodiments, without limitation, the AC governors **28** may periodically, continuously, randomly, combinations thereof, or the like, monitor AC current draw; the DC governor **30** may periodically, continuously, randomly, combinations thereof, or the like, monitor DC current draw; the thermal governor **32** may periodically, continuously, randomly, combinations thereof, or the like, monitor temperatures (e.g., by way of one or more sensors **24**) and thermal management subsystem **20** operations, respectively. Each unit **10** may comprise one, some, or all of the governors **28**, **30**, **32** in a same or different combination.

(24) Where the AC governor **28** determines that AC current draw is above a predetermined threshold, the AC governor **28** may initiate a current draw mitigation mode. The predetermined threshold may be between 70-99% of the service rated expected installed AC power, though any threshold may be utilized. The current draw mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** to rapidly, such as in a matter of seconds or less than 1 second, begin reducing illumination levels of the electronic displays **12**, such as by dimming the backlight. This may reduce or prevent nuisance tripping of circuit breakers, such as at the power overload prevention subsystem **18**. In this way, the current draw mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the current draw mitigation mode until AC current draw is below the predetermined threshold, such as by at least a margin to prevent continued movement between normal operation mode and current draw mitigation mode. Where the AC governor **28** determines that AC current draw is below the predetermined threshold, the AC governor **28** continue with normal operations.

(25) The AC governor **28**, control subsystem **14**, and/or power subsystem **16** may monitor current supplied, such as on a continual, periodic, and/or random basis. Where the current draw is below an

expected level, an electronic notification may be automatically generated and transmitted, such as by way of one or more network communication devices, to one or more remote electronic devices (e.g., computers, smart phones, tablets, servers, etc.). The network communication devices and transmission may be made by way of one or more internets, intranets, cellular networks, combinations thereof, or the like.

(26) Expected power supply levels, including current supply levels, may vary based on unit **10** configuration, such as size, number, and/or type of the electronic displays **12** installed, anticipated driving levels for the electronic displays **12**, other equipment **22** installed, expected ambient conditions, combinations thereof, and the like. Such power consumption levels may vary from approximately 200 watts to 5000+ watts, though any power level may be expected. Such expected current may vary from less than 2 amps to over 25 amps, though any current level may be expected. Circuit breakers may be configured to trip at less than 5 amps to over 30 amps, though any threshold may be utilized. These are provided by way of non-limiting example.

(27) Actual power supplied may vary based on power source **26** type and/or operational fluctuations, economic issues (e.g., operator unable or unwilling to pay for certain power supply at peak times, by way of non-limiting example), combinations thereof, or the like.

(28) Where the DC governor **30** determines that DC current demand is above a predetermined threshold, the DC governor **30** may initiate a power demand mitigation mode. The predetermined threshold may be reflective of limits of one or more installed power supplies, such as forming part of the power subsystem **16**. The predetermined threshold may be between 80-100% of the limits of the one or more installed power supplies, though any amount may be utilized. The power demand mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** to at a relatively moderate pace, such as in a matter of seconds (e.g., between 3-60 seconds) or minutes, begin reducing illumination levels of the electronic displays **12**, such as by dimming the backlight. This may reduce or prevent the one or more power supplies from reaching their maximum limit and turning off. In this way, the power demand mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the power demand mitigation mode until DC current demand is below the predetermined threshold, such as by at least a margin to prevent continued movement between normal operation mode and power demand mitigation mode. Where the DC governor **30** determines that DC current demand is below the predetermined threshold, the DC governor **30** may continue with normal operations.

(29) If the one or more power supplies reach their current limit, they may be configured to automatically shut off and be restarted.

(30) Where the thermal governor **32** determines that the thermal management subsystem **20** is operating at a predetermined capacity threshold and internal temperatures, such as determined by the one or more sensors **24**, are above a predetermined temperature threshold, the thermal governor **32** may initiate a temperature rise mitigation mode. Stated another way, the thermal management subsystem **20** may be performing at a maximum level for heat removal or some threshold thereof. For example, without limitation, the predetermined capacity threshold may be between 80-100% of the capacity of the thermal management subsystem **20** (e.g., fans operating at 80-100% of maximum speed), though any amount may be utilized. The predetermined temperature threshold may be the same or different for each sensor **24**, such as based on location and/or tolerance of local components. In exemplary embodiments, without limitation, the internal temperature condition for entering the temperature rise mitigation mode may be met where any one of the internal temperatures are above the predetermined temperature threshold and/or the respective predetermined temperature threshold for the sensor **24**. In other exemplary embodiments, without limitation, a plurality, or all, of the internal temperatures are above the predetermined temperature threshold and/or the respective predetermined temperature threshold for the sensor **24** before the temperature conditions for entering the temperature rise mitigation mode are met.

(31) The temperature rise mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** at a relatively slow pace, such as in a matter of minutes (e.g., between 5-60 minutes), begin reducing illumination levels of the electronic displays **12**, such as by dimming the backlight, such as down to a zero level. This may reduce or prevent damage to temperature sensitive components of the unit **10**. In this way, the temperature rise mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the temperature rise mitigation mode until the internal temperatures and/or thermal management subsystem **20** capacity is below the respective predetermined thresholds, such as by at least a margin to prevent continued movement between normal operation mode and temperature rise mitigation mode. Where the thermal governor **32** determines that internal temperatures and/or thermal management subsystem **20** capacity are below the predetermined threshold, the thermal governor **32** may continue with normal operations.

(32) These governors **28**, **30**, and **32** may be particularly important for preventing shut down of the units **10** and/or related components, such as the thermal management subsystems **20** which, if shut down, particularly in a moment of existing extreme operating conditions, may trigger a rapid rise in internal temperatures and comprise of the units **10**. Thus, the governors **28**, **30**, and **32** and related operations may serve to maintain operations of the unit **10**, such as the thermal management subsystem **20**, and/or minimize disruption to user experiences.

(33) The various predetermined thresholds and/or criteria for the governors **28**, **30**, and/or **32** may be set and/or varied by programming, such as by way of receipt of authenticated change commands from one or more remote electronic devices.

(34) The governors **28**, **30**, and/or **32** may operate independently from one another, in exemplary embodiments without limitation. The analysis undertaken by each of the governors **28**, **30**, and/or **32** may be performed in parallel and/or in any sequence.

(35) The integrated power governing features may alternatively, or additionally, permit the reduction in power supplies, such as of the power subsystem **16**, for a unit **10** and/or reduce power consumption for a unit **10**, among other benefits. This may allow provided power supplies, such as of the power subsystem **16**, to operate closer to maximum capacity, whereby increased efficiencies are generally found, thereby increasing operational efficiency of the unit **10** and reducing costs of manufacture, among other benefits.

(36) As illustrated with particular regard to FIG. **4**, requests for operational changes may be prioritized and/or centralized for disposition. For example, without limitation, operational change requests made by the governors **28**, **30**, and/or **32** may be centrally dispositioned, such as at the control subsystem **14**. Alternatively, or additionally, the governors **28**, **30**, and/or **32** may be in intercommunication and/or communication with a particular one of the governors **28**, **30**, and/or **32** which may be designated for arbitrating and/or prioritizing incoming requests. Where multiple requests for operational changes are received, such as from the different governors **28**, **30**, and/or **32** at the control subsystem **14**, the control subsystem **14** or other dispositioning unit (e.g., designated one of the governors **28**, **30**, and/or **32**) may be configured to prioritize the requests based on an amount of operational adjustment requested. For example, the largest operational change may be prioritized over the other requests. In exemplary embodiments, this may require acting on the request which results in the largest power and/or electronic display **12** illumination level reduction.

(37) The remaining requests may be discarded, ignored, and/or placed on hold. The remaining requests may be acted on subsequently, such as when the particular request being acted on is no longer valid, and/or the requests may be discarded as the various requests become no longer valid. For example, without limitation, the highest prioritized request may comprise the largest illumination level reduction which may be sufficient to request the lower-level requests. As another example, without limitation, once the highest prioritized request is performed for a period of time, the request may drop off due to sufficient changes to operational conditions that render the request

no longer needed, and the control subsystem **14**, for example, may move to a secondary request, tertiary request, etc. which remain valid based on updated operating conditions.

(38) While three specific governors **28**, **30**, and/or **32** may be utilized in exemplary embodiments, one or more additional governors **31** may optionally be utilized in place of, or in addition to, the governors **28**, **30**, and/or **32**. For example, without limitation, one of the additional governors **31** may be configured to trigger electronic display **12** dimming and/or other power consumption reduction efforts where one or more operational failures in the thermal management system **20** are detected. For example, without limitation, where one or more fan failures are detected. Such operational failures may be detected through lack of signal response, lack of power supply, combinations thereof, or the like. In other exemplary embodiments, without limitation, one of the additional governors **31** may be configured to trigger electronic display **12** dimming and/or other power consumption reduction efforts where external power supply changes or ceases, such as from one or more solar panels, utility power supplies, wind turbines, combinations thereof, or the like.

(39) Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention.

(40) Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, combinations thereof, and the like, configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may comprise personal computers, smartphones, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein. The electronic devices, including but not necessarily limited to the electronic storage devices, databases, controllers, or the like, may comprise and/or be configured to hold solely non-transitory signals.

Claims

1. A display unit for prioritizing requested operational updates, said display unit comprising: an electronic display; a thermal management subsystem; one or more sensors; and a control subsystem in electronic communication with the electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices with software instructions, which when executed, configure one or more processors to: operate the display unit based on operational parameters which are specific to readings from the one or more sensors; receive requests to update the operational parameters for the display unit; for each of the requests, determine a change in value to an associated one or ones of the operational parameters or an expected change in energy consumption for the display unit if the respective request were implemented; and prioritize the requests for implementation based on the change in value or the expected change in energy consumption.

2. The display unit of claim 1 wherein: the software instructions, when executed, configure the one or more processors to: for each of the requests, determine the change in value to the associated one or ones of the operational parameters or the expected change in energy consumption for the display

unit if the respective request were implemented on an absolute basis.

3. The display unit of claim 1 wherein: the software instructions, when executed, configure the one or more processors to: implement the requests in accordance with the prioritization.

4. The display unit of claim 1 wherein: the software instructions, when executed, configure the one or more processors to: discard or ignore all of the requests aside from a highest prioritized one of the requests.

5. The display unit of claim 1 wherein: the operational parameters in the requests comprise illumination setpoints for the electronic display.

6. The display unit of claim 5 wherein: the one or more sensors comprise at least one ambient light sensor; and the illumination setpoints are specific to ambient light readings from the at least one ambient light sensor.

7. The display unit of claim 5 wherein: the one or more sensors comprise at least one temperature light sensor; and the illumination setpoints are specific to temperature readings from the at least one temperature sensor.

8. The display unit of claim 5 wherein: the electronic display comprises a directly backlit liquid crystal display (LCD).

9. The display unit of claim 1 wherein: the operational parameters in the requests comprise setpoints for the thermal management subsystem.

10. The display unit of claim 9 wherein: the one or more sensors comprise at least one temperature sensor; and the setpoints are specific to temperature readings from the at least one temperature sensor.

11. The display unit of claim 10 wherein: the thermal management subsystem comprises one or more fans; and the setpoints comprise fan speeds levels.

12. The display unit of claim 11 wherein: the one or more fans comprise a plurality of fans; and the setpoints are different for at least some of the fans.

13. The display unit of claim 10 wherein: the at least one temperature sensor comprises a plurality of temperature sensors spaced apart within the display unit; and the setpoints are different for at least some of the temperature sensors.

14. The display unit of claim 1 wherein: the control subsystem comprises at least one of: an alternating current (AC) governor, a direct current (DC) governor, and a thermal governor; and the software instructions, when executed, configure the one or more processors to: assign each of the requests to a respective one of the governors for execution.

15. The display unit of claim 1 wherein: the software instructions, when executed, configure the one or more processors to: receive readings from the one or more sensors; and operate the display unit in accordance with the readings and the operational parameters.

16. A display unit for prioritizing requested operational updates, said display unit comprising: an electronic display; a thermal management subsystem; one or more sensors; and a control subsystem in electronic communication with the electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices with software instructions, which when executed, configure one or more processors to: operate the display unit based on operational parameters which are specific to readings from the one or more sensors; receive requests to update the operational parameters for the display unit; prioritize the requests for implementation; and discard or ignore the requests aside from a highest prioritized one of the requests.

17. A display unit for prioritizing requested operational updates, said display unit comprising: an electronic display; a thermal management subsystem; one or more sensors; and a control subsystem in electronic communication with the electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices with software instructions, which when executed, configure one or more processors to: operate the display unit based on operational parameters which are specific to

readings from the one or more sensors; receive requests to update the operational parameters for the display unit, wherein the operational parameters in the requests comprise illumination setpoints for the electronic display or setpoints for the thermal management subsystem; and prioritize the requests for implementation by magnitude of proposed change.
